Statistics: Preliminary Ph.D. Course

Day 1: Probability Theory

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Today's Session

We will follow the structure in Hansen's Book and then relate to Measure Theory

- **Outcomes and Events**: Definitions and Relation to Set Theory. The Partitioning Theorem: Statement and Proof. Examples.
- The **Probability Function**: Axioms of Probability. Properties of the Probability Function: Statements and Proofs. Examples.
- **Conditional Probability**: Definition. **Independence**: Definition. Examples.
- The **Law of Total Probability**: Statement and Proof. **Bayes Rule**: Statement and Proof. Examples.
- Relation to **Measure Theory**. Sigma Fields: Definition, Proofs and Examples.

Outcomes and Events

Definitions

- ***Definition 1.1***: An **outcome**, denoted by ω, is a specific result.
- ***Definition 1.2***: The **sample space**, denoted by Ω, is the set of all possible outcomes.
- ***Definition 1.3***: An **event** is a set A of outcomes in Ω.

Examples.

Set-theoretic manipulations are useful in describing events.

- ***Definition 1.4***: For events A and B:
  1. A is a **subset** of B, written $A \subseteq B$, if $\omega \in A \implies \omega \in B$.
  2. Two events are **equal**, written $A = B$, if $A \subseteq B$ and $B \subseteq A$.
  3. The event with no outcomes $\emptyset = \{\}$ is called the null or **empty set**.

Outcomes and Events

Definitions

- ***Definition 1.4 (cont.)***: For events A and B:

  4. The **union** of two events is the set $(A \cup B) := \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}$.
  5. The **intersection** $(A \cap B)$ of two events is the set $(A \cap B) := \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}$.
  6. The **complement** of A is the set $A^c := \{\omega \in \Omega : \omega \notin A\}$.
  7. The **set difference** of A with respect to B is the set $(A \setminus B) := \{\omega \in \Omega : \omega \in A, \omega \notin B\}$.
  8. The events A and B are **disjoint** if they have no outcomes in common: $(A \cap B) = \emptyset$.
  9. The events $A_1, A_2, \ldots$ are a **partition** of Ω if they are mutually disjoint and their union is Ω, this is:
  $$(A_i \cap A_j) = \emptyset \quad \forall i \neq j \quad \text{and} \quad \bigcup_n A_n = \Omega$$

Events satisfy the rules of set operations.

Theorem 1.1 (Partitioning Theorem)

Let $\{B_1, B_2, \dots\}$ be a partition of Ω. Then, for any event A it holds that:

$$A = \bigcup_i (A \cap B_i)$$

where the sets $(A \cap B_i)$ are mutually disjoint.

Proof.

The Probability Function

Definition

- ***Definition 1.5***: A **probability function** $\mathbb{P}$, is a function that assigns a numerical value to events and satisfies the following **Axioms of Probability**:

  1. $\mathbb{P}(A) \geq 0$
  2. $\mathbb{P}(\Omega) = 1$
  3. If $A_1, A_2, \ldots$ are disjoint, then:

  $$\mathbb{P}\left(\bigcup_n A_n\right) = \sum_n \mathbb{P}(A_n)$$

  The third axiom imposes considerable structure.

Examples.

The Probability Function

Properties

Theorem 1.2 (Properties of the Probability Function)

For events A and B,

1. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
2. $\mathbb{P}(\emptyset) = 0$
3. $\mathbb{P}(A) \leq 1$
4. **(Monotone Probability Inequality)** If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.
5. **(Inclusion-Exclusion Principle)** $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
6. **(Boole's Inequality)** $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$

Proof. *(Hint: Use the axioms of probability).*

Motivating Example: Take two events A and B, where A is defined as "*Passing the Econometrics I midterm exam*" and B as "*Study econometrics 12 hours a day*". We might be interested in the question (*and believe me, you are*): Does B affect the likelihood of A?

- ***Definition 1.6***: For any two events A and B, where $\mathbb{P}(B) > 0$, the **conditional probability** of A given B is:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Examples.

Independence

Definition

- ***Definition 1.7***: The events A and B are statistically independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Examples.

Theorem 1.3

If A and B are independent with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A|B) = \mathbb{P}(A) \quad \text{and} \quad \mathbb{P}(B|A) = \mathbb{P}(B)$$

Proof. *(Hint: Use the definitions of conditional probability and independence).*

Independent $\neq$ Disjoint

Law of Total Probability

Theorem

Theorem 1.4 (Law of Total Probability)

Let $\{B_1, B_2, \ldots\}$ be a partition of Ω and let $\mathbb{P}(B_i) > 0 \;\; \forall \; i$, then

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A|B_i)\mathbb{P}(B_i)$$

Proof. *(Hint: Use the partitioning theorem, the third axiom, and the definition of conditional probability).*

Example.

Bayes Rule

Theorem

Theorem 1.5 (Bayes Rule)

If $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|A^c)\mathbb{P}(A^c)}$$

Proof. *(Hint: Use the definition of conditional probability and the law of total probability).*

Example.

Relation to Measure Theory

The Bigger Picture (maybe too big)

- The definition of the Probability Function $\mathbb{P}$ given before in Definition 1.5 is **technically** incomplete.
- The problem is the type of events, A. When there are uncountable infinity of events it is necessary to restrict the set of allowable events to exclude pathological cases.
- Hansen: **“This is a technicality which has little impact on practical econometrics”**.
- But you will need to know for the Econometrics I course. Hence, I will walk you through some of these technicalities so you do not panic that much at the beginning.

- **Definition 1.8** (Probability Function II): A **Probability Function** $\mathbb{P}$, is a set-valued function from a σ-field $\mathcal{F}$ to the real line $\mathbb{R}$ which satisfies the Axioms of Probability:
  1. $\mathbb{P}(A) \geq 0$
  2. $\mathbb{P}(\Omega) = 1$
  3. If $A_1, A_2, \ldots$ are disjoint, then:

$$\mathbb{P}\left(\bigcup_n A_n\right) = \sum_n \mathbb{P}(A_n)$$

This restricts the domain of $\mathbb{P}$ to a class of sets $\mathcal{F}$, by excluding some problematic events (sets in Ω) for which probability is not defined. What is a σ-field?

Relation to Measure Theory

Definition of σ-field

- **Definition 1.9**: A **Sigma-Field** or **Sigma-Algebra** $\mathcal{F}$ is a collection (or family) of subsets of Ω that satisfies the following properties/axioms:

  1. $\Omega \in \mathcal{F}$.
  2. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
  3. If $A_1, A_2, \cdots \in \mathcal{F}$, then the **countable** union $\bigcup_n A_n \in \mathcal{F}$.

- We can thus prove the following:

  1. $\emptyset \in \mathcal{F}$
  2. $\bigcap_n A_n \in \mathcal{F}$

  Proof. *(Hint: Use the definition of σ-field and DeMorgan's Law).*

In general, a σ-field is closed under set operations, which means that the result of set operations applied to sets in $\mathcal{F}$ will also be in $\mathcal{F}$. The pair $(\Omega, \mathcal{F})$ is a **measurable space**, and the sets in $\mathcal{F}$ are **measurable sets**.

Relation to Measure Theory

Examples of σ-fields

- A σ-field can be generated from a finite collection of events. *Examples.*
- **Definition 1.10**: For a set A, its **Power Set** is the collection of sets $\mathcal{P}(A) := \{B : B \subseteq A\}$.

Examples.

- **Claim 1.1**: $\mathcal{P}(\Omega)$ is a σ-field on Ω. ***Proof.*** *(Hint: Use the definition of σ-field).*
- **Claim 1.2**: The intersection of σ-fields is a σ-field but the union of σ-fields is not necessarily a σ-field. ***Proof.*** *(Hint: Use the definition and a counter-example).*
- **Definition 1.11**: For a set Ω and a collection $\mathcal{H}$ of subsets of Ω, i.e., $\mathcal{H} \subseteq \mathcal{P}(\Omega)$, the **$\sigma$-field generated by $\mathcal{H}$**, denoted by $\sigma(\mathcal{H})$, is the intersection of all σ-fields on Ω that contain $\mathcal{H}$.
- $\sigma(\mathcal{H})$ is the **smallest σ-field** containing $\mathcal{H}$ (subcollection). *Examples.*

Relation to Measure Theory

Properties of $\sigma(\mathcal{H})$

Theorem 1.6 (Existence-Uniqueness of $\sigma(\mathcal{H})$)

Let Ω be a set and $\mathcal{H}$ a collection of subsets of Ω. $\sigma(\mathcal{H})$ exists and is unique.

Proof. *(Hint: Use the definition of $\sigma(\mathcal{H})$, Claim 1.2, and the definition of subset).*

- **Claim 1.3**: $\sigma(\sigma(\mathcal{H})) = \sigma(\mathcal{H})$. ***Proof.***
- **Claim 1.4**: $\mathcal{A} \subseteq \mathcal{B} \implies \sigma(\mathcal{A}) \subseteq \sigma(\mathcal{B})$. ***Proof.***
- **Claim 1.5**: $\mathcal{B} \subseteq \sigma(\mathcal{A}) \implies \sigma(\mathcal{B}) \subseteq \sigma(\mathcal{A}) = \sigma(A \cup B)$. ***Proof.***
- There is a very important example of a generated σ-field $\sigma(\mathcal{H})$ in probability theory: set $\Omega = \mathbb{R}$ and $\mathcal{H} = \mathcal{C}$, the collection of all open intervals on $\mathbb{R}$:

$$\mathcal{C} := \{(a, b) : \ a, b \in \mathbb{R} \text{ and } a < b\}$$

Relation to Measure Theory

Example: The Borel σ-field

- **Definition 1.12**: The **Borel σ-field** $\mathcal{B}$ (on $\mathbb{R}$) is the smallest σ-field on the real line $\mathbb{R}$ containing all open intervals (a, b): $\mathcal{B} = \sigma(\mathcal{C})$.
- Elements of $\mathcal{B}$ are called **Borel Sets**. It can be similarly defined on $\mathbb{R}^k$.
- **Claim 1.6**: Let $\mathcal{O}$ be the collection of all open sets on $\mathbb{R}$ and let $\mathcal{C}$ be the collection of all open intervals on $\mathbb{R}$. Then $\mathcal{B} = \sigma(\mathcal{O})$. ***Proof.*** *(Hint: Use the fact that any open interval I is an open set and that any open set U can be expressed as a countable union of open intervals: $U = \cup_n I_n$).*

- When there are an (uncountable) infinite number of events, it may not be possible to generate a σ-field through set operations, as there are pathological counter-examples.
- These counterexamples are difficult to characterize, are non-intuitive, and have **no practical implications for econometric practice**.
- The concept of σ-field is very abstract and technical. It gives **structure** to the sample space Ω so that we can safely assign probabilities (measures) to events (sets).

More on this the day after tomorrow.

Recommended Exercises

Hansen's Book and Class Notes

- Chapter 1, Exercises: 1.1-1.11, 1.18.
- Re-do all the Measure Theory proofs we covered.

References

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